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Jongho Kim

Sage Hall, 114 Feeney Way, Ithaca, NY 14850
Cornell Johnson Graduate School of Management

jk2352@cornell.edu
<https://jonghkim.github.io>

Education

- Cornell University, SC Johnson Graduate School of Management** Aug 2021–Present
Ph.D. in Quantitative Marketing
Minor Field: Applied Statistics
Committee: Vrinda Kadiyali (Chair), Emaad Manzoor, Omid Rafeian, Dan Kowal
- Korea Advanced Institute of Science and Technology** Feb 2015–Feb 2017
M.S. in Management Engineering, Information Systems
- Ulsan National Institute of Science and Technology** Mar 2010–Aug 2014
B.S. in Computer Science and Engineering, Minor in Finance and Accounting

Journal Publications

- Park, J. and **Kim, J.**, 2022. “A Data-Driven Exploration of the Race between Human Labor and Machines in the 21st Century,” *Communications of the ACM* [Manuscript] [SSRN] [Website] [Video] [Code]
- Choi, S., Kim, W. and **Kim, J.**, 2016. “Compression of Hamiltonian Matrix: Application to Spin-1/2 Heisenberg Square,” *AIP Advances* (**Editor’s Pick**) [Manuscript]

Working Papers

- Kim, J.** and Kadiyali, V., “Estimating Treatment Effects with Large Language Models: Evidence from San Diego’s Algorithmic Pricing Ban” (**Job Market Paper**)
- Kim, J.**, Kowal, D., and Ruppert, D., “Statistical Inference with Large Language Models: A Bayesian Framework for Measurement Errors”
- Kim, J.**, Anand, P., and Kadiyali, V., “The Effects of Rent Control on Renter Experience and Consumption”

Selected Conference and Seminar Presentations

- Kim, J.**, Kowal, D., and Ruppert, D., 2026, “Statistical Inference with Large Language Models: A Bayesian Framework for Measurement Errors,” *IO / Empirical Modeling Brown Bag*, Cornell University, Ithaca, NY, USA
- Kim, J.**, Anand, P., and Kadiyali, V., 2025, “The Effects of Rent Control on Consumer Reviews on Rental Housing,” *ISMS Marketing Science Conference*, Washington, D.C., USA

Park, J. and **Kim, J.**, 2018. “Leveraging Machine Learning to Reduce Racial Bias on Online Platforms: A Neural Machine Translation Approach,” *INFORMS Conference on Information Systems and Technology (CIST)*, Phoenix, Arizona, USA

Park, J. and **Kim, J.**, 2018. “Fixing Racial Discrimination through Analytics on Online Platforms: A Neural Machine Translation Approach,” *International Conference on Information Systems (ICIS)*, San Francisco, California, USA [Manuscript] [Poster]

Kim, J. and Park, J., 2017. “Does Facial Expression Matter Even Online? An Empirical Analysis of Facial Expression of Emotion and Crowdfunding Success,” *International Conference on Information Systems (ICIS)*, Seoul, Korea [Manuscript] [Poster]

Kim, J., Lee, M., Cho, D. and Lee, B., 2016. “Are All Spillovers Created Equal? The Impact of Blockbusters and the Composition of Backers in Online Crowdfunding,” *International Conference on Information Systems (ICIS)*, Dublin, Ireland [Manuscript] [Presentation]

Kim, J., Cho, D. and Lee, B., 2016. “The Mind Behind Crowdfunding: An Empirical Study of Speech Emotion in Fundraising Success,” *International Conference on Information Systems (ICIS)*, Dublin, Ireland (**Most Innovative Research-in-Progress Paper Runner-up**) [Manuscript] [Poster]

Park, J., **Kim, J.** and Lee, B., 2016. “Which Tasks Will Technology Take? A New Systematic Methodology to Measure Task Automation,” *International Conference on Information Systems (ICIS)*, Dublin, Ireland [Manuscript] [Poster]

Professional Experience

December & Company Asset Management, Seoul, South Korea Apr 2019–Jul 2021
Senior Researcher / Quantitative Portfolio Manager, Portfolio Research Team

- Develop Generative Models for Textual Factors to Predict and Explain Asset Returns based on SEC-10X (10-K, 10-Q, 10-KSB and 10-QSB)
- Implement a Sector Rotation Strategy through Cross-Sectional Variation over Data-driven Economic Cycles
- Construct Alternative Investment Strategies, especially for Commodity, REITs, Infrastructure and High Dividend-Paying ETFs
- Work as a Substitute for Mandatory Military Service

NICE Pricing and Information, Seoul, South Korea Aug 2016–Apr 2019
Quantitative Researcher, Financial Engineering Center

- Develop Non-Homogeneous Time Markov Chain Models (NHCTMC) for Forward Looking Probability of Default under IFRS9 based on Sparse Transition Matrix of Credit Ratings
- Estimate Maximum Likelihood Proxy Portfolio of Fixed-Income Portfolio via Return based Style Analysis (RBSA)
- Implement Implied Volatility Surface of Index Futures Options via Heston Stochastic Volatility Model
- Work as a Substitute for Mandatory Military Service

Digital Economics and Enterprise Research Lab, KAIST, Seoul, Korea May 2015–Feb 2017

- Empirical Research in Information Systems using Machine Learning and Econometrics (Advisor: Dr. Byungtae Lee)
- Accepted Three Conference Papers in ICIS 2016 and Awarded for the Most Innovative Research-in-Progress Paper Award (Runner-Up) in ICIS 2016
- Government & Industry Projects
 - Strategies for Online to Offline (O2O) Transport Services, The Korea Transport Institute (Advisor: Dr. Lee, B.), Sep 2016–Nov 2016
 - The Effect of Model-Driven Development (MDD) on Information Systems Maintenance, LG CNS (Advisor: Dr. Lee, B.), Sep 2015–Dec 2015

Computational Mathematical Science Lab, UNIST, Ulsan, Korea Sep 2011–Feb 2015

- Research and Study for Complex Network Theory and Numerical Methods for Differential Equations
- Published a Paper in AIP Advances (Compressed Representation of the Hamiltonian Matrix) and Selected as an Undergraduate Research Opportunity Program (Optimal Control Strategies of Lotka–Volterra Equation)

NewsJAM, Ulsan, South Korea Jun 2013–Feb 2014

Chief Technology Officer (CTO), Co-Founder

- Developed NLP engines for news analysis including text summarization, key sentence extraction, topic modeling, and text clustering [PyCon Korea 2016]
- Open SW Incubator (\$60,000 as award), National IT Industry Promotion Agency, 2013

Honors & Awards

AMA-Sheth Doctoral Consortium Fellow	2026
ISMS Doctoral Consortium Fellow	2025
FinTech@Cornell Student Fellowship	2022
Cornell SC Johnson College Doctoral Fellowship	2021–current
Best Student Paper Award, Post-ICIS 2017 KrAIS Research Workshop	2017
Best Graduate Student Paper Award, KAIST	2017
Most Innovative Research-in-Progress Paper Runner-Up, International Conference on Information Systems (ICIS), Dublin, Ireland	2016
Editor’s Picks from the American Institute of Physics, AIP Advances	2016
Scholarship for Academic Excellence among 2015 Entering Class, KAIST	2015
Excellence Prize (Top 1%, 3rd Place over 322 teams) at Hackathon for Big Data, Ministry of Science, ICT and Future Planning	2014
Dean’s List for Four Semesters, UNIST	2010, 2012–2014
Undergraduate Research Opportunity Program, Korea Foundation for Advancement of Science & Creativity	2013
National Natural Science and Engineering Scholarship, Korea Student Aid Foundation	2010–2014

Patents

Kim, J., et al., “Automated Trading System and Method,” Korean Patent 10-2433324, Assignee: December & Company Inc., 2022. (Application No. 10-2019-0117628, filed Sep 24, 2019)

Teaching

Instructor

Cornell University Fall 2025

BANA 5440 Introduction to Data Programming, MSBA

StudyPie Oct 2018–Apr 2019

Deep Learning Models for Financial Time Series Prediction [Link]

Guest Lecturer

KAIST FMBA – Introduction to FinTech (Prof. Lee, B.) Mar 2021, Mar 2023, Apr 2024

Investment Strategies in FinTech Industry [Link]

Korea Summer Session on Causal Inference (Prof. Park, J.) Jul 2021

Applications of Machine Learning in Marketing [Link]

KAIST MBA – Business Analytics (Prof. Oh, W.) Apr 2019

Intelligent Video Analytics with Deep Learning [Link]

KAIST IT Management Summer Research Workshop (Prof. Park, J.) Aug 2017

Machine Learning APIs and Web Scraping [Link]

Teaching Assistant

Cornell University

NCCT 5030/5031 Marketing Management Spring, Fall 2023

AEM 4435 / NBA 6390 Data-Driven Marketing Fall 2022

NCC 5030 Marketing Management Summer 2022

Invite-Only Workshops

Econometric Society, Africa Training Workshop in Econometrics, 2024

Services

Ad-hoc Reviewer

International Conference on Information Systems (ICIS), 2021

International Conference on Information Systems (ICIS), 2019

International Conference on Electronic Commerce (ICEC), 2017

Other Information

Work Authorization: Employment Authorization Document (EAD) Expected

Language: Korean (Native), English (Fluent)

Programming: Python, C/C++, Java, R, STATA, Matlab, SQL, JavaScript, HTML, CSS

References

Vrinda Kadiyali (Chair)

Nicholas H. Noyes Professor of Management
Professor of Marketing and Economics
SC Johnson Graduate School of Management
Cornell University
kadiyali@cornell.edu

Emaad Manzoor

Assistant Professor of Marketing
SC Johnson Graduate School of Management
Cornell University
emaadmanzoor@cornell.edu

Omid Rafieian

Assistant Professor of Marketing
Demir Sabanci Faculty Fellow
Cornell Tech and SC Johnson College of Business
Cornell University
or83@cornell.edu

Dan Kowal

Associate Professor of Statistics and Data Science
Bowers College of Computing & Information Science
Cornell University
dan.kowal@cornell.edu

Selected Research Abstracts

“Estimating Treatment Effects with Large Language Models: Evidence from San Diego’s Algorithmic Pricing Ban,” Kim, J. and Kadiyali, V. (Job Market Paper)

How can we use LLM outputs for causal inference when they are imperfect rather than ground truth? LLMs read constructs from text at scale, but they hallucinate mentions that are absent, omit ones that are present, and miscalibrate the values they report. In this study, the construct of interest is what a careful human reader would read from a text. We develop a semiparametric Bayesian framework that estimates its treatment effect by combining large-scale LLM outputs with a small labeled sample, correcting for the LLM’s hallucination, omission, and miscalibration. We apply it to San Diego’s 2025 algorithmic-pricing ban, which bars setting rents from competitors’ nonpublic data, across 58,293 Google Maps reviews of 1,686 apartments. Nine months after the ban, renters rate their apartments 0.15 points lower on a 5-point scale, counter to its pro-renter intent. In the text, satisfaction with rent price falls 0.26 while non-price services rise 0.37. The ban appears to raise pricing inefficiency and shift landlords from price to service competition, leaving renters worse off. As the DOJ–RealPage settlement takes the same regime nationwide, our findings inform how policymakers weigh its consequences.

“The Effects of Rent Control on Renter Experience and Consumption: Evidence from St. Paul’s 2022 Ordinance,” Kim, J., Anand, P., and Kadiyali, V.

How does housing affordability policy reshape consumer experience and grocery spending? We examine the impact of St. Paul’s 2022 rent control policy using 110,258 renter reviews from 2,285 apartments and grocery sales data from 123 stores in 18 months post-policy. We compare St. Paul (policy area) and surrounding spillover areas to comparable U.S. cities as controls. Renters rate their apartment experiences higher by 0.474 points (on a 5-point scale) in the policy area and by 0.157 points in the spillover areas. In the 18 months after policy implementation, no dimension of the multi-attribute service experience worsens, suggesting net satisfaction gains for renters in our time window. Grocery expenditure declines by 8% in the policy area and 3% in the spillover areas over the two years following implementation through different mechanisms: the exit of higher-income renters from the policy area and likely increased rents in the spillover areas. This reveals a policy tradeoff: improved renter satisfaction dampened grocery demand in affected neighborhoods. Our findings have implications for apartment managers to optimize service quality and consumer retention under price regulation; grocery retailers to recalibrate assortment and pricing for shifting neighborhood demographics; and policymakers to weigh the full consequences of rent control across renters, businesses, and spillover areas.

“Statistical Inference with Large Language Models: A Bayesian Framework for Measurement Errors,” Kim, J., Kowal, D., and Ruppert, D.

Large language models (LLMs) enable researchers to measure latent constructs from text at scale. We study treatment effect estimation using LLM-measured outputs as causal outcomes. Standard plug-in estimators can substantially underestimate uncertainty, even when point estimates are unbiased. An LLM’s errors are not independent. It misreads semantically similar units in the same direction, so the errors are correlated. This correlation is information, not nuisance, but its structure is unknown a priori. We develop a Bayesian framework that learns it from the embedding geometry with a Gaussian process, estimating it jointly with the calibration and the treatment effect from a small labeled sample and a large set of unlabeled LLM outputs. Estimating jointly, not in stages, lets the labeled and unlabeled data discipline each other, and the learned covariance of errors down-weights redundant reads. Together these restore valid inference and improve precision over methods that assume independent errors. In a randomized experiment we document that this dependence is real, survives fine-tuning, and escapes conventional clustering, and across simulations our framework holds valid coverage while improving efficiency. The result shows that the dependence among an LLM’s errors can be exploited rather than ignored.